HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** applications for manufacturing execution, integrating with production databases and replacing manual data entry workflows.
- Designed and implemented **REST APIs** for cross-system data exchange between manufacturing, ERP, and quality management platforms, ensuring secure data interfaces.
- Automated repetitive tasks and data validation using **Python scripting**, reducing manual data entry errors by an estimated **20%** across operational reporting processes.
- Created user-friendly **shop-floor interfaces** and **dashboards** displaying production progress and quality metrics, directly translating factory user requirements into functional software.
- Managed software releases through **Git version control**, conducted **code reviews** for new feature development, and wrote automated tests covering critical failure scenarios.
- Collaborated with the Quality team to document software requirements and provided evidence for **software validation**, managing controlled changes through Jira and Confluence.
- Containerized key applications using **Docker** and managed deployments to **AWS EC2** instances, ensuring consistent environments from development through production.

SKILLS

Languages – Python, Java, SQL, JavaScript, TypeScript

Frameworks/Libraries – Spring Boot, Flask, FastAPI, pandas, NumPy, scikit-learn

Databases – PostgreSQL, MySQL, Oracle, Relational Databases

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins

Tools/Platforms – Jira, Confluence, Splunk, Dynatrace, API Integration, ERP Systems, Quality Systems

Methodologies – Agile, Scrum, CI/CD, Code Review, Automated Testing

PROJECTS

Manufacturing Execution System Enhancement (HCL Technologies)

- Developed new features for a **Java Spring Boot**-based manufacturing execution system, focusing on improving traceability and reducing manual data input for inspection processes.
- Engineered **API integrations** between the MES and an enterprise **ERP system**, enabling seamless data flow for production order management and material tracking.
- Implemented automated data validation checks within the application, catching approximately **15%** of potential data integrity issues before they impacted downstream systems.

Production Data Dashboard & Automation (HCL Technologies)

- Built a **Python**-scripted data pipeline to aggregate production metrics from various factory systems, feeding into a centralized reporting database.
- Created interactive **dashboards** using data visualization tools to display real-time production status, capacity utilization, and key quality indicators for operations teams.
- Automated the generation of daily operational reports, saving an estimated **4 hours** of manual work per week for the production support team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)